

Chronic stress impairs autoinhibition in neurons of the locus coeruleus to increase asparagine endopeptidase activity

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This **fundamental** study explores a novel cellular mechanism underlying the degeneration of locus coeruleus neurons during chronic restraint stress. The evidence supporting the overexpression of LC neurons after chronic stress is **compelling**. However, to fully support the broad implications for LC degeneration and Alzheimer's disease, the study would benefit from stronger causal integration and validation in age-relevant models.

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Abstract

Impairments of locus coeruleus (LC) are implicated in anxiety/depression and Alzheimer's disease (AD). Increases in cytosolic noradrenaline (NA) concentration and MAO-A activity initiate the LC impairment through production of NA metabolite, 3,4-dihydroxyphenylglycolaldehyde (DOPEGAL), by MAO-A. However, how NA accumulates in soma/dendritic cytosol of LC neurons has never been addressed despite the fact that NA is virtually absent in cytosol while NA is produced exclusively in cytoplasmic vesicles from dopamine by dopamine- β -hydroxylase. Since re-uptake of autocrine-released NA following spike activity is the major source of NA accumulation, we investigated whether and how chronic stress can increase the spike activity accompanied by NA-autocrine. Overexcitation of LC neurons is normally prevented by the autoinhibition mediated by activation of α 2A-adrenergic receptor (AR)-coupled inwardly rectifying potassium-current (GIRK-I) with autocrine-released NA. Patch-clamp study revealed that NA-induced GIRK-I in LC neurons was decreased in chronic restraint stress (RS) mice while a similar decrease was gradually caused by repeated excitation. Chronic RS caused internalization of α 2A-ARs expressed in cell membrane in LC

neurons and decreased protein/mRNA levels of $\alpha 2A$ -ARs/GIRKs in membrane fraction. Subsequently, chronic RS increased the protein levels of MAO-A, DOPEGAL-induced asparagine endopeptidase (AEP) and tau N368. These results suggest that chronic RS-induced overexcitation due to the internalization of $\alpha 2A$ -ARs/GIRK is accompanied by $[Ca^{2+}]_i$ increases, subsequently increasing Ca^{2+} -dependent MAO-A activity and NA-autocrine. Thus, it is likely that internalization of $\alpha 2A$ -AR increased cytosolic NA, as reflected in AEP increases, by facilitating re-uptake of autocrine-released NA. The suppression of $\alpha 2A$ -AR internalization may have a translational potential for AD treatment.

Background

Locus coeruleus (LC) is implicated in anxiety and depression (Itoi and Sugimoto, 2010), and is often the first place where Alzheimer's disease (AD)-related pathology appears (Mather and Harley, 2016). Noradrenaline (NA) is known to inhibit amyloid-induced oxidative stress, and caspase activation in cortex/hippocampus (Counts and Mufson, 2010). Thus, the impairment of LC-noradrenergic system is potentially involved in the AD pathogenesis (Ross et al., 2015; Chen et al., 2022).

It was reported that cytosolic NA in the soma/dendrites of LC neurons is degraded by monoamine oxidase A (MAO-A) into 3,4-dihydroxyphenyl-glycolaldehyde (DOPEGAL) (Burke et al., 1999), which generates apoptotic free radicals (Burke et al., 1998) and activates asparagine endopeptidase (AEP) to produce A β and hyperphosphorylated tau, consequently leading to AD (Burke et al., 1999; Kang et al., 2020). However, there appears normally no NA except dopamine in the cytosol of LC neurons because NA is produced exclusively in cytoplasmic vesicles by dopamine- β -hydroxylase (DBH) (Potter and Axelrod, 1963) from dopamine which is taken up into cytoplasmic vesicles by vesicular monoamine transporter type 2 (VMAT2).

LC neurons release NA following action potential not only from the axon terminals but also from their cell bodies as autocrine (Huang et al., 2007), and autocrine-released NA activates $\alpha 2$ adrenergic receptor ($\alpha 2$ -AR) expressed on the soma/dendrites (Aoki et al., 1994), resulting in activation of $\alpha 2$ -AR-coupled inwardly rectifying K^+ (GIRK) channel (Arima et al., 1998) to cause autoinhibition (spike-afterhyperpolarization) (Williams et al., 1985). Thus, overexcitation of LC neurons is normally prevented by the autoinhibition.

As autocrine-released NA is taken up by NA-transporter (NAT) (Torres et al., 2003), uptaken-NA may be the only source of free NA to be metabolized by MAO-A in the cytosol of LC neurons. However, the amount of free NA to be metabolized by MAO-A in the cytosol is thought to be very small due to a much lower affinity of NA to MAO-A compared to VMAT2 (Costa, 2012). Therefore, it is not clear how free NA to be metabolized by MAO-A is increased in the soma/dendrites of LC neurons in AD pathogenesis. Because the passive NA-leakage occurs from cytoplasmic vesicles in a manner that achieves a dynamic equilibrium with active NA-storage into cytoplasmic vesicles by VMAT2 (Eisenhofer et al., 2004), increased NA-uptake by NAT in response to increased NA-autocrine following $[Ca^{2+}]_i$ increases or overexcitation may increase free NA by facilitating the passive NA-leakage from cytoplasmic vesicles. Furthermore, the affinity of MAO-A for NA may be modifiable as MAO-A activity is Ca^{2+} -dependent (Cao et al., 2007). Then, a question arises whether or not impairment of autoinhibition that causes overexcitation and $[Ca^{2+}]_i$ increases can consequently increase AEP activity through the production of DOPEGAL.

Neuronal activity in LC neurons is transiently increased by stress (Valentino and Foote, 1988) through activation of corticotropin releasing factor (CRF) receptor (Valentino et al., 1983; McCall et al., 2015) together with glutamate receptors (Valentino et al., 2001). However, even if firing activity is increased by synaptic activation (Barcomb et al., 2022), autoinhibition would occur to suppress overexcitation in LC neurons. Nevertheless, it is known that long-term overexcitation

due to chronic stress ultimately causes degeneration of LC neurons (Kitayama et al., 2004). Then, this raises an important question whether and how stress impairs autoinhibition to cause sustained overexcitation. The whole mechanisms for the impairment of LC would be revealed by solving these questions. Here we examined whether the autoinhibition mediated by $\alpha 2A$ -AR-coupled GIRK-I is impaired by overexcitation in LC neurons or chronic stress, to clarify the mechanisms for NA accumulation and/or MAO-A activity increase.

Results

Spike frequency adaptation in LC neurons is mediated by $\alpha 2$ -AR-coupled GIRK channel

LC neurons invariably display spike-frequency adaptation (Figure 1A, B). We have hypothesized that the autoinhibition would occur following respective spike firing and contribute to the generation of the spike-frequency adaptation and the afterhyperpolarization seen at the offset of current pulses.

When the cumulative number of spikes was plotted against the elapsed time from the onset of the current pulse every 50 ms, the cumulative spike number increased non-linearly in a saturation manner (Figure 1C). This non-linear relationship was fitted with a saturation curve defined by Monod equation (see Extended Methods and Materials) in order to quantify the spike-frequency adaptation. Because there was no significant difference in b between the two groups of spike trains, the profile of the normalized cumulative spike number vs the elapsed time remains the same regardless of firing frequency, suggesting that spike-frequency adaptation occurs in the same way.

The spike-frequency adaptation observed was almost completely abolished by a blocker of $\alpha 2$ -AR, atipamezole (Figure 1D, E), which subsequently largely decreased pulse-afterhyperpolarization (pulse-AHP) (Figure 1D, E, blue and red stars). The cumulative spike number increased non-linearly in a saturation manner in the control condition while it increased almost in a linear manner after atipamezole application (Figure 1F). In the relationship between a and b examined in LC neurons, atipamezole invariably and significantly shifted both a and b in a larger direction, and there was a significant difference in the relationship between a and b (Figure 1G).

Bath application of NA induced a GIRK-I in response to the ramp pulse (Figure 1H, red trace) whereas addition of atipamezole to NA almost completely abolished the NA-induced GIRK-I (Figure 1H, blue trace), showing that GIRK-I reversed at -97 mV. The amplitudes of the inward component of NA-induced GIRK-I were invariably larger than those of the outward component, consistent with the inwardly rectifying property of GIRK-I. Atipamezole significantly decreased the amplitudes of the inward and outward components (Figure 1I). Taken together, NA-autocrine is likely to be caused following respective spikes in the LC neuron and NA is accumulated around the LC neuron to activate $\alpha 2$ -AR-coupled GIRK channels, to cause spike-frequency adaptation by enhancing AHP.

Ca^{2+} -dependent rundown of NA-induced GIRK-I in LC neurons

In PC12 cells, heterologously expressed $\alpha 2A$ -ARs undergo internalization following agonist application (Taraviras et al., 2002). Such internalization of $\alpha 2A$ -ARs would result in an abolishment of GIRK-I or desensitization given the functional coupling between $\alpha 2A$ -AR and GIRK as in LC noradrenergic neurons. However, in LC neurons, NA-induced GIRK-I showed almost no rundown/desensitization after repeating the ramp pulse 20 times in the persistent presence of NA (Figure supplement 1A-D). Then, we next examined whether GIRK-I shows rundown under the

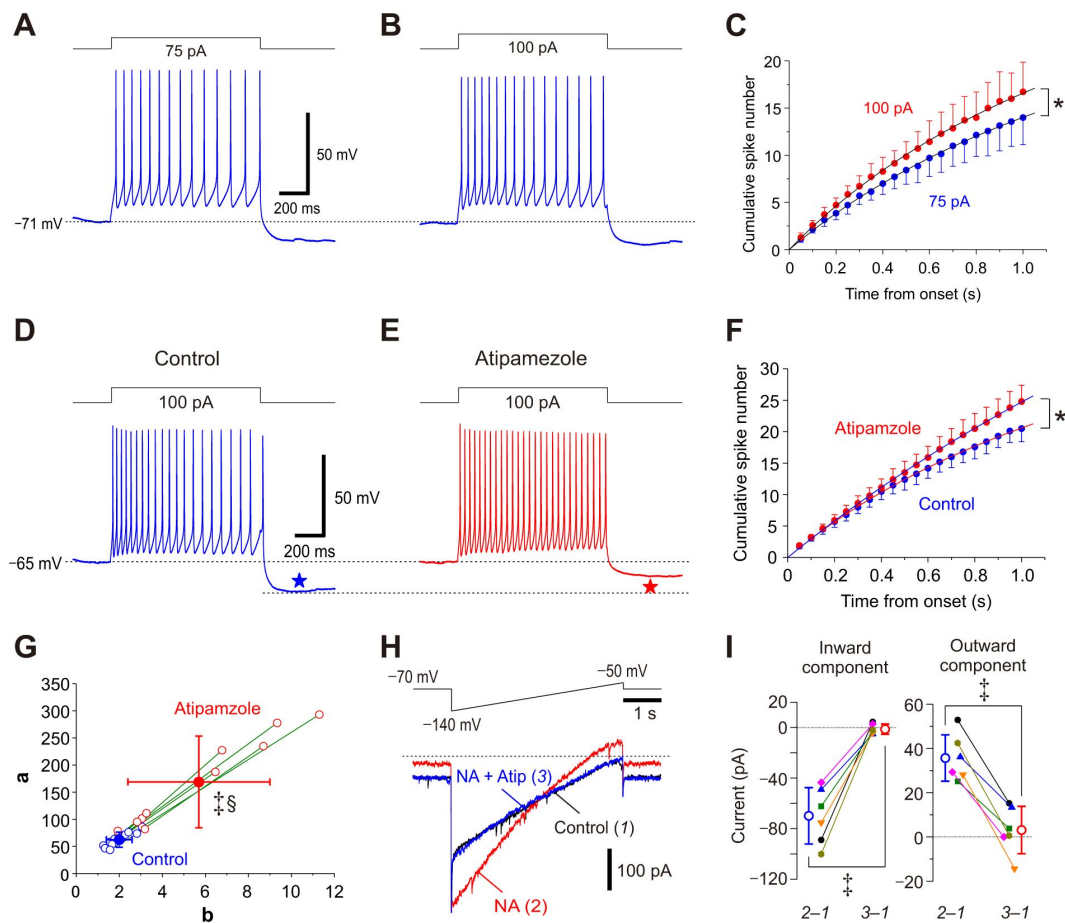


Figure 1.

Atipamezole abolishes spike-frequency adaptation and blocks NA-induced GIRK-I in LC neurons.

(A, B) Sample traces of spike trains induced in an LC neuron by injection of current pulses at 75 and 100 pA. Note the spike-frequency adaptation and pulse-afterhyperpolarization (pulse-AHP).

(C) Plotting of the cumulative spike numbers vs the elapsed time from the onset of the current pulses every 50 ms at 75 pA (blue circles) or 100 pA (red circles) ($n = 7$). Two-way RM ANOVA, $*p < 0.001$. The saturation level (**a**) and the half saturation constant (**b**) were measured by fitting the saturation function defined as $y = (a * x) / (b + x)$ to the data points. The values of **a** at 75 and 100 pA current pulses were 44.6 ± 16.2 and 52.5 ± 12.3 , respectively (paired t -test, $p = 0.010$). The values of **b** at 75 and 100 pA current pulses were 2.1 ± 0.7 and 2.1 ± 0.4 , respectively (paired t -test, $p = 0.823$).

(D, E) Sample traces of spike trains induced in an LC neuron evoked by current pulses at 100 pA under control condition and after application of 10 μ M atipamezole. Note that atipamezole suppresses spike-frequency adaptation and pulse-AHP (blue and red stars).

(F) Plotting of the cumulative spike numbers vs the elapsed time during the current pulses every 50 ms before and after application of atipamezole (blue and red circles, respectively) ($n = 10$). Two-way RM ANOVA, $*p < 0.001$.

(G) Plotting of **a** against **b**, which were measured by fitting the saturation function to the data points in F. Both the values of **a** and **b** obtained by curve fitting to the data points after application of atipamezole (open and filled red circles) were significantly shifted to larger values than those obtained under control condition (open and filled blue circles) (paired t -test, **a** and **b**, $^{\dagger}p = 0.003$ and $^{\dagger}p = 0.002$, respectively), and there was a significant difference in the relationship between **a** and **b** (Wilk's lambda, $^{\S}p = 0.002$) ($n = 10$).

(H) Top, Voltage command pulse. Lower panel, Superimposed current traces obtained under control condition (1), after application of 100 μ M NA (2), and after application of 10 μ M atipamezole (Atip) in addition to NA (3). The amplitudes of inward and outward components in NA-induced GIRK-I were -70 ± 22 and 36 ± 11 pA, respectively (paired t -test, $^{\dagger}p = 0.005$, $n = 6$).

(I) A graph showing the effects of atipamezole on amplitudes of the inward and outward components of NA-induced GIRK-I at -130 mV and -60 mV, obtained by subtraction of currents recorded under control condition from those recorded after application of NA (2-1) and from those recorded after application of NA and atipamezole (3-1). Inward component, paired t -test, $^{\dagger}p < 0.001$; outward component, paired t -test, $^{\dagger}p < 0.001$ ($n = 6$).

condition of “increased excitability” or increased $[Ca^{2+}]_i$, because acute/chronic stress increases the excitability of LC neurons (Borodovitsyna et al., 2018 [↗](#); Campos-Lira et al., 2018 [↗](#)) and various channels/receptors are known to be internalized in a Ca^{2+} -dependent manner (Beattie et al., 2000 [↗](#); Aziz et al., 2012 [↗](#); Zaccor et al., 2020 [↗](#)).

To examine the Ca^{2+} -dependency of the rundown of GIRK-I, the respective ramp pulses to evoke GIRK-I were preceded by various pulse trains which can induce Ca^{2+} currents differentially (compare Figure supplement 1F and **Figure 2B** [↗](#)). After examining various pulse protocols (Figures supplement 1 and 2), the best protocol to induce a prominent rundown of GIRK-I was as follows (shown in **Figure 2B** [↗](#)). First, bath application of NA induced GIRK-I (**Figure 2A** [↗](#), red trace; 2) that showed the inwardly rectifying profile as revealed by subtraction of the control response from that obtained after NA application (**Figure 2D** [↗](#), red trace; 2–1).

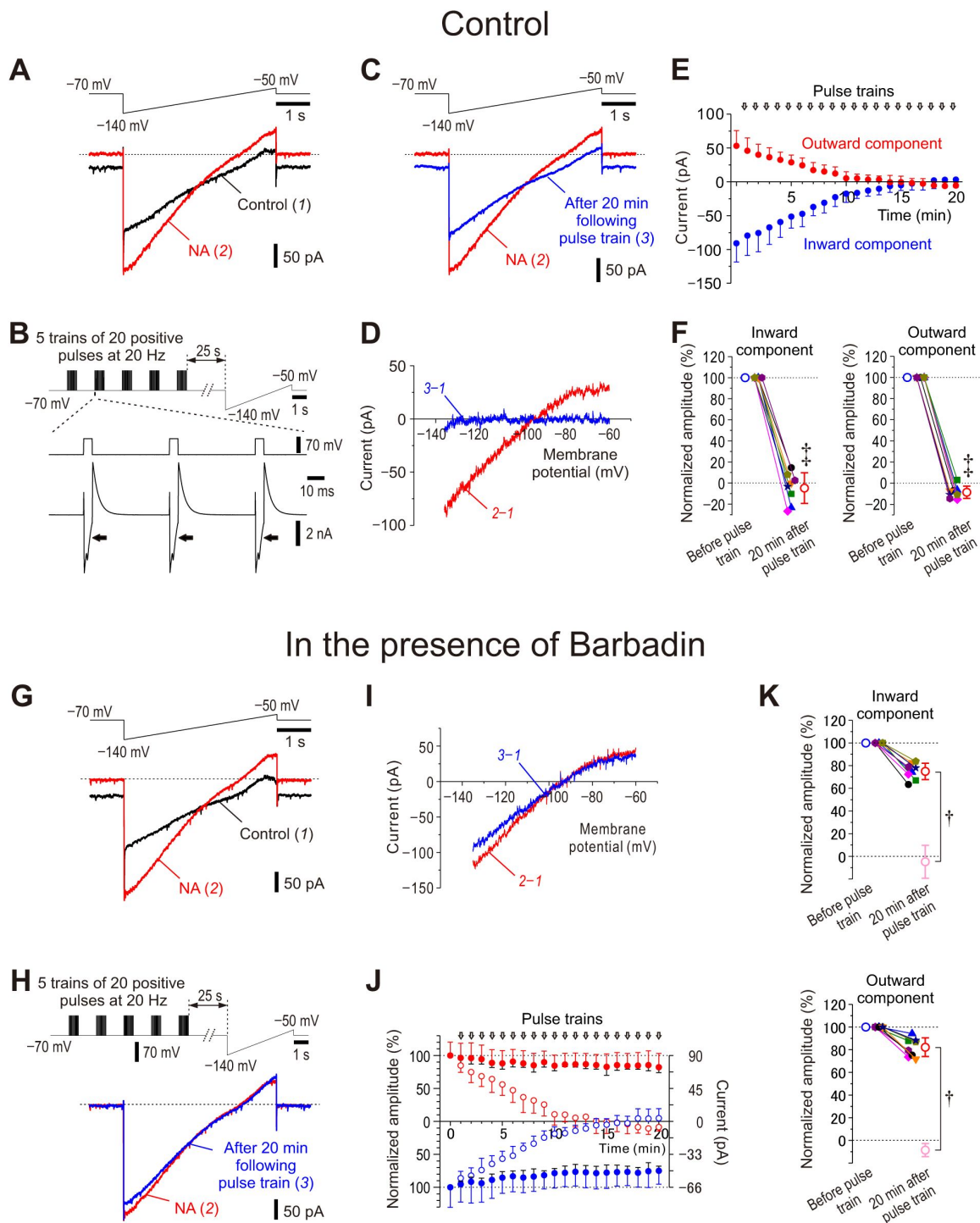


Figure 2.

Ca²⁺-dependent rundown of NA-induced GIRK-I and its suppression by barbadin.

(A) Upper panel, Ramp command pulse. Lower panel, Superimposed current traces obtained under control condition (1) and after application of 100 μ M NA for 5 min (2).

(B) Upper panel, A combined command pulse applied every minute; five trains of 20 positive pulses (5 ms duration to 0 mV at 20 Hz) at an interval of 2 s in the presence of extracellular 30 mM TEA and intracellular 0.2 mM EGTA, which were followed by the ramp pulse after an interval of 25 s. Lower panel, representative Ca²⁺ currents in response to positive pulses in the presence of NA and 30 mM TEA.

(C) Upper panel, Ramp command pulse. Lower panel, Superimposed current traces obtained after application of NA for 5 min (2) and in response to application of the 20th combined pulse in the presence of NA and TEA (3).

(D) The I-V relationship of NA-induced GIRK-I obtained by subtraction of currents recorded under control condition from that recorded after application of NA for 5 min (red trace, 2–1) and that obtained by subtraction of the control current from that recorded in response to application of the 20th combined pulse in the presence of NA and TEA (blue trace, 3–1).

(E) Plotting of amplitudes of inward components at –130 mV (blue circles) and outward components at –60 mV (red circles) against time.

(F) Plotting of normalized amplitudes of inward components at –130 mV and outward components at –60 mV before and 20 min after application of positive pulse trains in the presence of NA and TEA. The amplitudes of inward components at –130 mV and those of outward components at –60 mV were normalized by those recorded in the presence of NA before applying positive pulse trains. Inward component, paired *t*-test, [‡]*p* < 0.001; outward component, paired *t*-test, [‡]*p* < 0.001 (*n* = 8).

(G) Upper panel, Ramp command pulse. Lower panel, Superimposed current traces obtained under control condition (1) and after application of 100 μ M NA for 5 min (2) in LC neurons dialyzed with 100 μ M barbadin (β -arrestin/AP2 blocker).

(H) Upper panel, A combined command pulse applied every minute; five trains of 20 positive pulses (5 ms duration to 0 mV at 20 Hz) at an inter-train interval of 2 s in the presence of extracellular 30 mM TEA and intracellular 0.2 mM EGTA, which were followed by the ramp pulse after an interval of 25 s. Lower panel, Superimposed current traces obtained after application of NA for 5 min (2) and in response to application of the 20th combined pulse in the presence of NA and TEA (3) in LC neurons dialyzed with barbadin.

(I) The I-V relationship of NA-induced GIRK-I obtained by subtraction of currents recorded under control condition from that recorded after application of NA for 5 min (red trace, 2–1) and that obtained by subtraction of the control current from that recorded in response to application of the 20th combined pulse in the presence of NA and TEA (blue trace, 3–1).

(J) Plotting of normalized amplitudes (left vertical axis) of inward components at –130 mV (filled blue circles) and outward components at –60 mV (filled red circles) against time (*n* = 8). Amplitudes of GIRK-I were normalized to the values obtained before applying pulse trains. Right vertical axis refers to the original amplitudes. Open blue and red circles represent the normalized amplitudes of GIRK-I shown in E.

(K) Plotting of normalized amplitudes of inward components at –130 mV and outward components at –60 mV before and 20 min after application of positive pulse trains in the presence of NA and TEA (*n* = 8). The amplitudes of inward components at –130 mV and those of outward components at –60 mV were normalized to those recorded in the

presence of NA before applying positive pulse trains. Open light pink circles represent results obtained under the control condition in F. Inward component, unpaired *t*-test, **p* < 0.001; outward component, unpaired *t*-test, **p* < 0.001.

Subsequently, the combined command pulse (a set of 5 trains of 20 pulses followed by a ramp pulse; **Figure 2B**, upper trace) was applied every minute. The positive pulse trains evoked large inward Ca^{2+} currents in the presence of extracellular 30 mM TEA (**Figure 2B**, lower traces) whereas similar trains of the same positive pulse hardly induced Ca^{2+} currents when applied in the absence of TEA (Figure supplement 1F, lower trace). With repetition of the combined command pulse 20 times, the NA-induced GIRK-I (**Figure 2C**, red trace; 2) was decreased to almost the same current as the control response (**Figure 2C**, blue trace; 3). Indeed, no GIRK-I remained after 20 times repetition of the combined command pulse as revealed by the subtraction between trace 1 and trace 3 (**Figure 2D**, blue trace; 3–1). Both the amplitudes of the inward and outward components of the NA-induced GIRK-I promptly decreased to zero along with repetition of the combined command pulse (**Figure 2E**, blue and red circles; **Figure 2F**). Thus, it is likely that NA-induced GIRK-I displays a marked rundown in a Ca^{2+} -dependent manner. Furthermore, we confirmed that these rundowns were completely dependent on the number and application timing of the command pulse in the presence or absence of TEA (compare **Figure 2** with Figures supplement 1 and 2). Depending on the absence or presence TEA, the single application of a set of 10 trains of 20 pulses differentially (slightly or moderately, respectively) induced the rundown of NA-induced GIRK-I (compare Figure supplement 1E–J and Figure supplement 2A–E). By contrast, the repetitive application of a single train of 20 positive pulses in the presence of TEA markedly facilitated the rundown of NA-induced GIRK-I (Figure supplement 2F–K). The relationship between the number of spikes (i.e., positive pulse) and $[\text{Ca}^{2+}]_i$ was obtained by measuring Ca^{2+} transients using fura-2 (Figure supplement 3).

To investigate the effects of the rundown of GIRK-I on the spike-frequency adaptation, spike trains were examined before and after the voltage-clamp experiment (**Figure 2A–F**). After the rundown of GIRK-I was observed in the voltage-clamp experiment, the spike-frequency adaptation was absent in the current clamp experiment, and the pulse-ADP was largely reduced (Figure supplement 4A–D). The cumulative spike number that increased non-linearly in a saturation manner before rundown of GIRK-I increased almost in a linear manner after rundown of GIRK-I (Figure supplement 4E, F). Thus, the rundown of GIRK-I invariably resulted in an abolishment of spike-frequency adaptation, similar to the effects of atipamezole on spike-frequency adaptation (**Figure 1**). Therefore, it is likely that Ca^{2+} increases can cause the rundown of NA-induced GIRK-I and almost abolished the spike-frequency adaptation, leading to an increase in the excitability of LC neurons. These observations suggest that $[\text{Ca}^{2+}]_i$ increases cause rundown of NA-induced GIRK-I to suppress autoinhibition, leading to excitability increases that cause further increases in $[\text{Ca}^{2+}]_i$ to further cause a rundown of the GIRK-I.

It was also examined whether the rundown of NA-induced GIRK-I can be caused solely by $[\text{Ca}^{2+}]_i$ increases without activation of $\alpha 2\text{-AR}$. We found that NA-induced GIRK-I subjected to the combined command pulse 20 times in the presence of atipamezole was significantly restored after washout of atipamezole in the presence of NA (Figure supplement 5A–D). This observation indicates that GIRK-I were not downregulated solely by $[\text{Ca}^{2+}]_i$ increases without activation of $\alpha 2\text{-AR}$, clearly revealing that both the activation of $\alpha 2\text{-AR}$ and $[\text{Ca}^{2+}]_i$ increases are necessary to cause the downregulation of GIRK-I. In agreement with this result, the spike-frequency adaptation reflecting the autoinhibition was also preserved in spite of $[\text{Ca}^{2+}]_i$ increases as long as activation of $\alpha 2\text{-AR}$ is prevented (Figures supplement 5E–J). We further examined whether the rundown of NA-induced GIRK-I can be prevented by the presence of a GIRK channel blocker, tertiapin-Q. GIRK channel blockade did not prevent the rundown of NA-induced GIRK currents following $[\text{Ca}^{2+}]_i$ increases (Figure supplement 6A–H). In agreement with this result, the spike-frequency adaptation was not preserved by with the blockade of GIRK channels during $[\text{Ca}^{2+}]_i$ increases (Figure

supplement 6I-N). These findings showed that $\alpha 2$ -AR blockade but not GIRK blockade prevented the rundown of NA-induced GIRK-I following $[Ca^{2+}]_i$ increases. Taken together, it is suggested that the rundown of NA-induced GIRK-I may be brought about by the internalization of $\alpha 2$ -AR in a Ca^{2+} -dependent manner, regardless of opening or closing of GIRK channels.

β -Arrestin is involved in the rundown of NA-induced GIRK-I

We further examined whether the internalization of $\alpha 2$ -AR is involved in the rundown of NA-induced GIRK-I or not. As β -arrestin/AP2 complex is known to be involved in the agonist-dependent internalization of GPCR (DeGraff et al., 1999 [↗](#); Cottingham et al., 2011 [↗](#)), we examined the effects of barbadin, which prevents AP2 from binding with β -arrestin by forming barbadin/AP2 complex, on the rundown of NA-induced GIRK-I. With the patch pipettes filled with barbadin-including internal solution, it was examined whether NA-induced GIRK-I displays rundown in response to the same combined command pulse (**Figure 2H** [↗](#), upper trace) as that in the control experiment (**Figure 2B** [↗](#), upper trace). The NA-induced GIRK-I (**Figure 2G-I** [↗](#), red traces) was hardly decreased (**Figure 2H, I** [↗](#), blue traces) after repeating the combined pulse 20 times, in contrast to the control rundown (**Figure 2C, D** [↗](#)). Barbadin largely suppressed the rundown of the both components of GIRK-I (**Figure 2J, K** [↗](#)). These results suggest that the rundown of NA-induced GIRK-I is caused by the internalization of $\alpha 2$ -AR-coupled GIRK channels.

Restraint stress alters $\alpha 2A$ -AR expression in LC neurons

We next investigated whether the expression of $\alpha 2A$ -AR is altered in LC neurons by applying restraint stress (RS) to mice. After mice were subjected to daily 12-h RS for 3 days followed by non-stress condition for 2 days, the immunoreactivity for $\alpha 2A$ -ARs (see Figure supplement 7) seen along the membrane of TH-positive LC neurons was apparently lower compared to the control (**Figure 3A, B** [↗](#)). Consistent with this observation, Western blot analysis revealed significantly lower protein levels of $\alpha 2A$ -ARs in LC neurons obtained after 3-day RS compared to the control (**Figure 3C** [↗](#)). Especially when $\alpha 2A$ -ARs expression was examined in the membrane fraction identified using Na^+/K^+ -ATPase, the immunoreactivity for $\alpha 2A$ -ARs expression was apparently lower in LC neurons of RS mice than in those of the control mice (**Figure 3D, E** [↗](#); *, compare the rightmost lower panels). Consistent with this observation, Western blot analysis revealed a significantly lower protein level of $\alpha 2A$ -ARs in the membrane fraction of LC neurons of RS mice compared to the control whereas there was no significant difference in the cytosol between the control and RS mice (**Figure 3F, G** [↗](#)). Thus, the internalization of $\alpha 2A$ -ARs is not necessarily accompanied by increases in cytosolic levels of $\alpha 2A$ -ARs because the internalized receptors are either recycled or degraded (Koenig and Ikeda, 1996 [↗](#); Takei et al., 1996 [↗](#)).

RS reduces NA-induced GIRK-I

As 3-day RS caused the internalization or downregulation of $\alpha 2A$ -ARs, we next examined if the daily 12-h RS for up to 5 days can differentially cause reductions of NA-induced GIRK-I in LC neurons. NA-induced GIRK-I decreased with an increase in the RS period (**Figure 4A-E** [↗](#)). The amplitudes of inward and outward components of GIRK-I decreased with the increase in the period of RS (**Figure 4F, G** [↗](#)), in a way that is described by a saturation function (red interrupted lines). The amplitudes of inward and outward components of GIRK-I in 1-day mice were significantly smaller than those of the control mice, and those in 3-day RS mice were significantly smaller than those in 1-day RS mice while no significant difference in those between 3-day and 5-day RS mice (**Figure 4F, G** [↗](#)). Thus, the downregulation of GIRK-I was increased as RS period is increased, but in a saturation manner.

Quantitative PCR analysis revealed that LC neurons expressed much higher mRNA level of $\alpha 2A$ -AR than that of $\alpha 2C$ -AR (**Figure 4H** [↗](#)) and also expressed higher mRNA level of GIRK1 compared to GIRK2 while GIRK3 was hardly expressed (**Figure 4I** [↗](#)). Consistent with the significant decreases in NA-induced GIRK-I in the mice subjected to 3-day RS, mRNA levels of $\alpha 2A$ -AR, GIRK1 and GIRK2

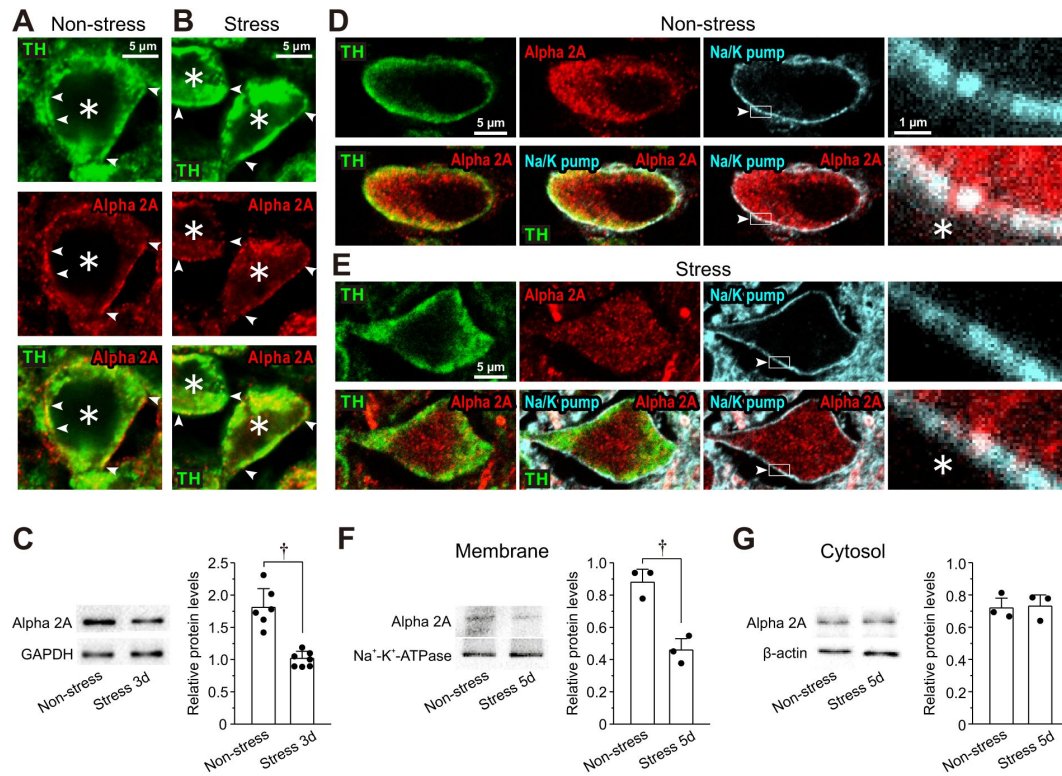


Figure 3.

RS reduces expression of α2A-ARs in LC Neurons.

(A, B) Confocal images of LC neurons showing immunoreactivities for TH and α2A-ARs, together with a merged one in non-stress and 3-day RS mice. Arrowheads indicate the membrane regions of TH-positive neurons, along which α2A-ARs were differentially expressed between the control and the RS mice. Asterisks indicate LC neurons.

(C) Western blotting analyses showing the expression of α2A-ARs in non-stress and 3-day RS mice ($n = 6$ and 7 , respectively). Unpaired t -test, $^{\dagger}p < 0.001$.

(D, E) Confocal images of LC neurons in non-stress and RS mice. Upper panels from left to right showing the respective immunoreactivity for TH, α2A-ARs (with rabbit anti-α2A-AR) and Na⁺-K⁺-pump and an enlarged image of the region enclosed with a rectangle (arrowheads) in its immediate left panel. Lower panels from left to right showing the merged image of TH and α2A-ARs, that of TH, α2A-ARs and Na⁺-K⁺-pump, that of α2A-ARs and Na⁺-K⁺-pump and an enlarged image of the region enclosed with a rectangle in its immediate left panel.

(F, G) Western blotting analyses showing expressions of α2A-ARs and Na⁺-K⁺-pump in membrane fraction and those of α2A-ARs and β-actin in cytosol fraction in non-stress and 5-day RS mice ($n = 3$ and 3 samples, respectively). Each sample represents the analysis result in the LC tissues obtained from 2-3 mice. Membrane; unpaired t -test, $^{\dagger}p = 0.016$. Cytosol; unpaired t -test, $^{\dagger}p = 0.083$.

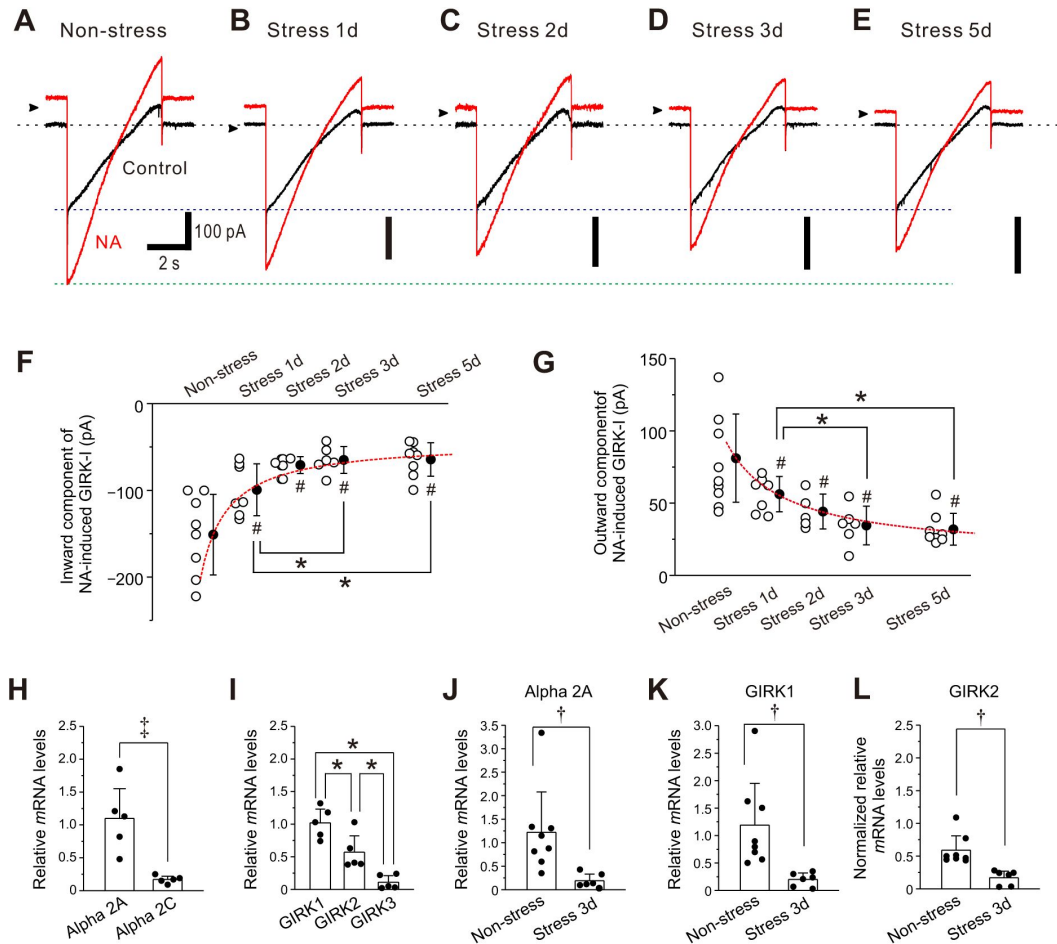


Figure 4.

RS reduces NA-induced GIRK-I in LC neurons.

(A-E) Representative traces of NA-induced GIRK-I obtained from LC neurons in non-stress and 1-day, 2-day, 3-day, and 5-day RS mice.

(F, G) Amplitudes of inward and outward components of NA-induced GIRK-I obtained from LC neurons in non-stress and 1-day, 2-day, 3-day, and 5-day RS mice ($n = 8, 7, 5, 6$ and 8 , respectively) decreased with the increase in the period of RS, in a way that can be described by a saturation function (red interrupted lines). The saturation level ($a + b$) and the half saturation constant (c) were determined by fitting the saturation function, defined as $y = a + (b \cdot x) / (c + x)$, to the data points. The values of a , b and c for the inward component of GIRK-I were -151.1 , 108 , and 0.9 , respectively, and those for the outward component of GIRK-I were 81.1 , -67 , and 1.6 , respectively. Inward component: one-way ANOVA, $p < 0.001$, post hoc fisher's PLSD, 1-day; $\#p < 0.001$ vs Non-stress and $*p < 0.05$ vs 3-day and 5-day, 2-day; $\#p < 0.001$ vs Non-stress, 3-day; $\#p < 0.001$ vs Non-stress, 5-day; $\#p < 0.001$ vs Non-stress. Outward component: one-way ANOVA, $p < 0.001$, post hoc fisher's PLSD, 1-day; $p = 0.004$ vs Non-stress and $*p < 0.05$ vs 3-day and 5-day, 2-day; $p = 0.015$ vs Non-stress, 3-day; $p < 0.001$ vs Non-stress, 5-day; $p < 0.001$ vs Non-stress.

(H) Relative expressions of $\alpha 2A$ and $\alpha 2C$ mRNAs, normalized to GAPDH in LC neurons ($n = 5$). Paired t -test, $^\dagger p = 0.014$.

(I) Relative expressions of GIRK1, GIRK2, and GIRK3 mRNAs, normalized to GAPDH in LC neurons ($n = 5$). One-way RM ANOVA, $p < 0.001$, post hoc fisher's PLSD; $*p = 0.008$ for GIRK1 vs GIRK2, $*p < 0.001$ for GIRK1 vs GIRK3, $*p = 0.008$ for GIRK2 vs GIRK3.

(J, K) Relative expressions of $\alpha 2A$ and GIRK1 mRNAs, respectively, normalized to GAPDH in LC neurons in non-stress mice ($n = 8$), and 3-day RS mice ($n = 6$). $\alpha 2A$ -AR: unpaired t -test, $^\dagger p = 0.020$; GIRK1: unpaired t -test, $^\dagger p = 0.013$.

(L) Normalized relative expressions of GIRK2 mRNA in LC neurons in non-stress mice ($n = 8$) and 3-day RS mice ($n = 6$), normalized to the ratio of the mean value of the relative expressions of GIRK1 mRNA to that of GIRK2 mRNA in LC neurons in non-stress mice (I). Unpaired t -test, $^\dagger p = 0.037$.

in mice subjected to 3-day RS were significantly smaller than those of the control mice (**Figure 4J-L**). Thus, RS downregulated the expression/production of $\alpha 2A$ -AR-coupled GIRK1 prominently in LC neurons.

Histological and biochemical changes in LC neurons after 5-day RS

Next, immunohistochemical experiments demonstrated that 5-day RS appeared to slightly increase the expressions of TH and MAO-A in LC neurons (Figure supplement 8A). Furthermore, AEP expression also appeared to increase in cytosol after 5-day RS compared to the control (Figure supplement 8B). Because TH catalyzes the rate limiting step in the synthesis of catecholamines, if the RS enhances TH activity, free NA concentration in LC neurons would be directly and greatly increased. Therefore, we first carefully examined whether TH is really upregulated in RS mice or not by performing Western blot analysis. No significant increase in the protein level of TH was observed in 5-day RS mice compared to the control (**Figure 5A**, upper panel; **Figure 5B**). This result clearly indicates that TH activity was not significantly upregulated in 5-day RS mice, in spite of the TH immunohistochemical experiment. We further examined whether protein levels of MAO-A and AEP are increased or not by 5-day RS. Western blot analysis revealed significant increases in the protein levels of MAO-A (**Figure 5A**, lower panel; **Figure 5C**) and of pro-AEP and active AEP (**Figure 5D-F**) in LC neurons after 5-day RS compared to the control. Western blot analysis also revealed a significant increase in the expression of tau and tau N368 fragment after 5-day RS (**Figure 5G-I**). Thus, an increased activity of active AEP caused a production of tau N368 fragments as observed at 37 and 35 kDa (**Figure 5G**), which is consistent with a previous report (Zhang et al., 2014). Relative protein levels of the tau N368 fragments (**Figure 5I**) were the mean values obtained from the two band intensities, which may reflect the two N-terminal fragments (upper and lower bands) generated by AEP cleavage at Asn-368 of the 4 and 3 repeat-tau isoforms, respectively (Goedert et al., 1989). Finally, Y-maze test revealed that 5-day RS decreased spatial memory (Figure supplement 9).

Discussion

We have found that LC neurons display spike-frequency adaptation which is caused by the autoinhibition mediated by $\alpha 2A$ -AR-coupled GIRK-I following respective spikes (**Figure 1**) and that $\alpha 2A$ -AR-coupled GIRK-I showed Ca^{2+} -dependent rundown due to the internalization of $\alpha 2A$ -AR (**Figure 2**, Figure S5A-D and Figure S6E-H), leading to the abolishment of spike-frequency adaptation and the excitability increase (Figure supplement 4). Immunohistochemical examination and Western blot analysis revealed that $\alpha 2A$ -AR expression in cell membrane was decreased following RS (**Figures 3** and **4J**). Consistently, NA-induced GIRK-I was decreased in LC neurons with an increase in the period of RS (**Figure 4A-G**). Following chronic RS, protein levels of MAO-A, pro-/active-AEP, and tau and tau N368 fragment were significantly increased (Figure supplement 8A, B and **Figure 5A, C-I**).

Based on these results, we propose a novel mechanism for how chronic RS impairs autoinhibition and increases cytosolic free NA to be metabolized by enhanced Ca^{2+} -dependent MAO-A activity, to consequently increase AEP activity and produce tau N368 protein (**Figure 6**). In non-stress mice, autocrine released NA is slowly taken up by NAT after binding to $\alpha 2$ -ARs to cause autoinhibition (**Figure 6A**), while in chronic RS mice, autocrine-released NA may be directly and rapidly taken up by NAT due to the absence (internalization) of $\alpha 2$ -ARs (**Figure 6B**). Such increased rate of re-uptake of NA by NAT would increase free NA concentration by facilitating the leakage of NA from cytoplasmic vesicles in the cell body, which would be metabolized by upregulated MAO-A activity to DOPEGAL, as revealed by the increased activity of AEP in chronic RS mice (**Figure 6B**). It may be further necessary to perform enzymatic assay to better demonstrate the AEP and MAO-A activities.

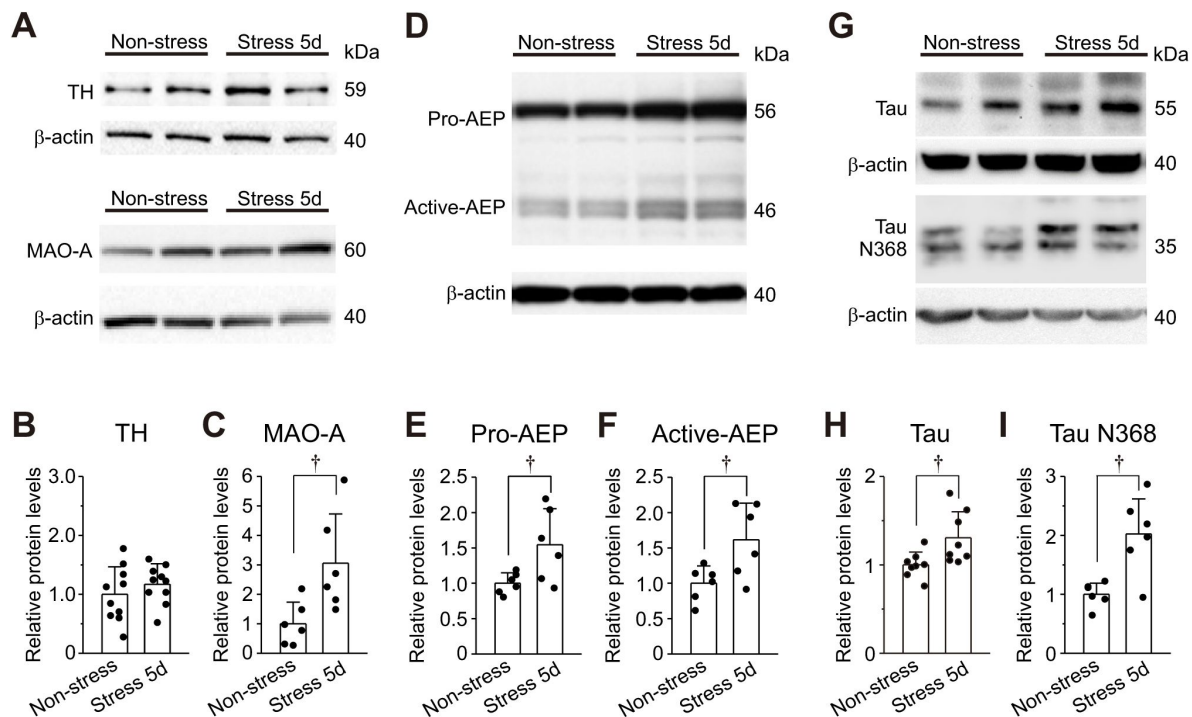


Figure 5.

Expressions of TH, MAO-A, DBH, AEP and tau proteins in LC neurons.

(A) Western blotting of TH and MAO-A in LC neurons obtained from non-stress and 5-day RS mice.

(B, C) Western blot analysis revealing no significant change in TH expression ($n = 10$ samples) but a significant increase in MAO-A expression ($n = 6$ samples) in 5-day RS group compared to non-stress group. Unpaired t -test, $^{\dagger}p = 0.029$.

(D) Western blotting of pro-AEP and active AEP in LC neurons obtained from non-stress and 5-day RS mice.

(E, F) Western blot analysis revealing significant increases in pro-AEP and active AEP ($n = 6$ and $n = 6$ samples, respectively) in 5-day RS group compared to non-stress group. Unpaired t -test, $^{\dagger}p = 0.034$ (E) and $^{\dagger}p = 0.047$ (F).

(G) Western blotting of tau and tau N368 fragment in LC neurons obtained from non-stress and 5-day RS mice. Unpaired t -test, $^{\dagger}p = 0.025$.

(H, I) Western blot analyses revealing a significant increase in tau and tau N368 fragment in LC neurons ($n = 8$, and $n = 6$, respectively) in 5-day RS group compared to non-stress group. Each sample represents the analysis result in the LC tissues obtained from 2-3 mice. Unpaired t -test, $^{\dagger}p < 0.05$.

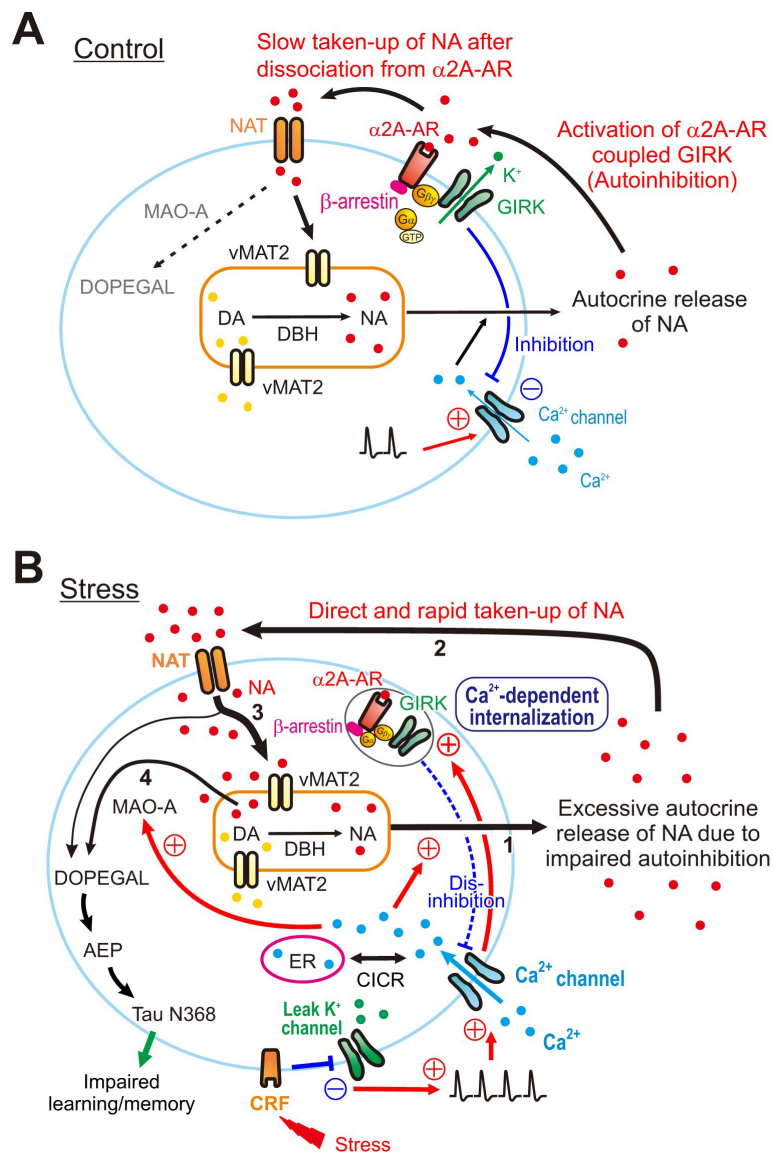


Figure 6.

Presumed cellular mechanisms for the degeneration of LC by stress.

(A, B) Differential free concentrations of NA to be metabolized by MAO-A into DOPEGAL between control and stress conditions: Under the control condition **(A)**, NA in the cytosol is mostly taken up into the cytoplasmic vesicles by VMAT2 (thick arrow 3), rather than being directly metabolized by MAO-A into DOPEGAL (thin interrupted arrow). Vesicular NA is released from cell bodies as autocrine following $[Ca^{2+}]_i$ increases caused by action potentials, and the released NA activates $\alpha 2A$ -AR-coupled GIRK channels, causing autoinhibition. Subsequently, the autocrine released NA is slowly taken up into the cytosol of LC neurons by NAT after dissociation from $\alpha 2A$ -ARs. Under the stress condition **(B)**, an activation of CRF receptors in LC neurons by stress inhibits leak K^+ channels and increases firing activities in LC neurons, subsequently causing a larger $[Ca^{2+}]_i$ increase together with Ca^{2+} -induced Ca^{2+} release (CICR). Impairment of autoinhibition due to Ca^{2+} dependent internalization of $\alpha 2A$ -ARs-coupled GIRK channels leads to the persistent excitation in LC neurons, which enhances autocrine release of NA (thick arrow 1). Subsequently, the excessively autocrine released NA is taken up directly and rapidly by NAT into the cytosol without binding to $\alpha 2A$ -ARs (thick arrow 2). Such a facilitation of re-uptake of NA by NAT would increase active NA storage into cytoplasmic vesicles by VMAT2 (thick arrow 3), while the rate of NA leakage from cytoplasmic vesicles would also increase (arrow 4) due to a dynamic equilibrium in cytoplasmic vesicles between active NA storage into cytoplasmic vesicles and passive NA leakage from cytoplasmic vesicles. Subsequently, such an increase in the rate of NA leakage would result in an increase in a MAO-A metabolite, DOPEGAL and AEP, leading to a production of cleaved tau N368 fragment and an impairment of learning/memory.

Spike frequency adaptation

LC neurons display spike-frequency adaptation. The mechanism for the spike-frequency adaptation may be different among neurons while M currents are usually responsible for the adaptation in hippocampal pyramidal neurons (Aiken et al., 1995 [↗](#)). In LC neurons, the spike-frequency adaptation was mediated by $\alpha 2$ A-AR-coupled GIRK-I as the spike-frequency adaptation was abolished by atipamezole (**Figure 1D-G** [↗](#)) and by causing the rundown of GIRK-I (Figure supplement 4). These results suggest that autocrine NA release following respective spikes activated $\alpha 2$ A-ARs expressed in the soma/dendrites of LC neurons, and subsequently activated $\alpha 2$ A-AR-coupled GIRK-I to cause autoinhibition as spike-frequency adaptation. Thus, spike-frequency adaptation clearly reflects NA-autocrine.

Rundown of $\alpha 2$ A-AR-coupled GIRK-I

$\alpha 2$ A-AR-coupled GIRK-I displayed Ca^{2+} -dependent rundown in the presence of agonist. The degree and time course of rundown of $\alpha 2$ A-AR-coupled GIRK-I were dependent on the number and timing of positive pulses applied under voltage-clamp condition to cause Ca^{2+} currents (**Figure 2A-F** [↗](#) and Figures supplement 1E-J, 2, 3). Given the Ca^{2+} -dependency of the rundown of autoinhibition, the excitability of LC neurons would increase as LC neurons were excited more such as under RS condition, resulting in a vicious cycle between excitability increase accompanied by $[\text{Ca}^{2+}]_i$ increase and rundown of autoinhibition/ Ca^{2+} -dependent internalization of $\alpha 2$ A-ARs.

Internalization of $\alpha 2$ A-ARs

Activation of GPCR by agonist binding usually causes the formation of GPCR/ β -arrestin complexes. Subsequently, the GPCR/ β -arrestin complex is assembled through the interaction between β -arrestin and AP2 in clathrin-coated pits (CCPs), which is formed by the association of AP2 with phosphatidylinositol 4,5-bisphosphate (PIP2) at the plasma membrane, leading to endocytosis of GPCRs (Pierce and Lefkowitz, 2001 [↗](#)). PIP2 is locally produced from the membrane phosphatidylinositol (PI) by PI4/PIP5 kinases in a Ca^{2+} -dependent manner (Sarmiento et al., 2014 [↗](#)). Barbadin prevented the stable formation of sufficient GPCR/ β -arrestin/AP2 complexes in CCPs as barbadin prevents AP2 from binding with β -arrestin by forming barbadin/AP2 complex, thereby hampering GPCR internalization (Beautrait et al., 2017 [↗](#)). In the present study, intracellular barbadin effectively suppressed the rundown of GIRK-I (**Figure 2G-K** [↗](#)), indicating that the rundown of GIRK-I was caused by the internalization of $\alpha 2$ A-ARs in LC neurons.

Chronic RS increases the activities of MAO-A and pro-/active-AEP in the cell body

Chronic cold stress reduced $\alpha 2$ -AR mediated inhibition of LC neurons (Jedema et al., 2008 [↗](#)), and chronic social stress downregulated $\alpha 2$ -ARs in the LC (Flugge, 1996 [↗](#)). Our present results were consistent with those previous findings: Following RS, $\alpha 2$ A-ARs expression was decreased due to the internalization of $\alpha 2$ A-ARs (**Figure 3** [↗](#)), together with the decreases in NA-induced GIRK-I (**Figure 4A-G** [↗](#)) which mediates the autoinhibition. As long as RS persists, the impairment of autoinhibition leads to the persistent overexcitation in LC neurons. This would further increase NA-autocrine, because respective spikes were accompanied by NA-autocrine as reflected in spike-frequency adaptation or autoinhibition.

As extracellular NA is mostly taken up into cytosol by NAT (Torres et al., 2003 [↗](#)), it is likely that under chronic RS the increased autocrine-released NA is directly and more rapidly taken up by NAT due to the absence of $\alpha 2$ A-ARs (**Figure 6B** [↗](#)). Such an increase in the rate of NA uptake by NAT is similar to the effects of uptake inhibitor of transmitter ligand on the activity of receptors. After re-uptake of NA by NAT, a large portion of NA in the cytosol is restored in cytoplasmic vesicles by VMAT2 while only a small portion of NA can escape from VMAT2 to be metabolized by

MAO-A (Eisenhofer et al., 1992 [↗](#)), because VMAT2 has a much higher affinity for NA than MAO-A (Bloom, 2006 [↗](#); Costa, 2012 [↗](#)). However, there is an additional component of NA to be metabolized by MAO-A, as NA can leak from cytoplasmic vesicles into cytosol (Eisenhofer et al., 2004 [↗](#)). When re-uptake of NA by NAT is facilitated due to the internalization of $\alpha 2$ -ARs in response to NA-autocrine, active NA storage into cytoplasmic vesicles by VMAT2 would increase. Then, the rate of NA leakage from cytoplasmic vesicles would also increase because passive NA leakage from cytoplasmic vesicles and active NA storage into cytoplasmic vesicles by VMAT2 are in a dynamic equilibrium in cytoplasmic vesicles (Eisenhofer et al., 2004 [↗](#)). Subsequently, such increases in the rate of NA leakage would result in increases in a MAO-A metabolite, DOPEGAL (Goldstein, 2021 [↗](#)) (**Figure 6B** [↗](#)). Indeed, it has previously been demonstrated that a partial inhibition of NAT in human subjects reduced the original level of a derivative of DOPEGAL (3,4-dihydroxyphenylglycol; DHPG) in CSF by more than 50% (Kielbasa et al., 2015 [↗](#)). Thus, free NA to be metabolized by MAO-A would be increased when autocrine-released NA is taken up by NAT. On the other hand, it is also known that the decreased VMAT2 levels, generated in VMAT2 heterozygote mice, increased the vulnerability of nigral dopamine neurons to MPTP (Takahashi et al., 1997 [↗](#)). Thus, it is conceivable that NA leakage for free NA accumulation is influenced by both NAT or VMAT2.

Therefore, under the chronic RS condition, more NA leaked from cytoplasmic vesicles is likely to be metabolized by MAO-A, which activity is enhanced Ca^{2+} -dependently and by chronic RS (Cao et al., 2007 [↗](#)) (**Figures 5A, C** [↗](#) and **6B** [↗](#)), to DOPEGAL compared to the control condition, as revealed by the increased protein levels of pro-AEP and active-AEP (Figure supplement 8B and **Figure 5D-F** [↗](#)). Thus, re-uptaken NA by NAT may largely be degraded to DOPEGAL by the increased activity of MAO-A. Subsequently, DOPEGAL converts pro-AEP into active-AEP to produce tau N368 protein (**Figure 5G-I** [↗](#)).

It has never been addressed how DOPEGAL production was caused in the cell body of LC neurons while the DOPEGAL production in the cell bodies of LC neurons is well established in AD patients (Burke et al., 1999 [↗](#); Kang et al., 2020 [↗](#)) and in transgenic AD model mice (Kang et al., 2020 [↗](#)). We demonstrated for the first time that 5-day RS can impair the autoinhibition by causing the internalization of $\alpha 2$ A-AR and consequently induces active-AEP to produce tau N368 fragment. Indeed, 5-day RS induced spatial memory impairment-like responses in mice when assessed using Y-maze test (Figure supplement 9) while anxiety-like behavior was also induced in mice after 3-day RS when assessed using an elevated plus maze test (figure not shown), both suggesting an impairment of LC neurons. It is likely that, in addition to NAT inhibitor, the suppression of the internalization of $\alpha 2$ A-AR has a strong translational potential for developing drugs that prevent the degeneration of LC.

Limitations of this study

One may argue against the usage of 3-week-old juvenile mice to study the neurodegenerative disorder which is normally found in aged ones. However, a stringent autoinhibition together with potent NAT activity were previously demonstrated in adult rats using intracellular recordings (Williams et al., 1985 [↗](#)). Thus, regardless of the age of mice, LC neurons showed the autoinhibition that is mediated by the activity of $\alpha 2$ A-AR-coupled GIRK channels. We also confirmed that spike-frequency adaptation reflecting autoinhibition can be induced in adult mice as prominently as in juvenile mice regardless of sexes (Figure supplement 10), although we did not investigate the sex differences in RS effects (Leistner and Menke, 2020 [↗](#)). The downregulation of autoinhibition by the internalization of $\alpha 2$ A-AR is agonist- and Ca^{2+} -dependent regardless of the age of mice. However, it is not known if there is any difference in the susceptibility of $\alpha 2$ A-AR internalization to RS between juvenile and aged mice. Thus, a further study may be necessary to examine if autoinhibition in LC neurons is more easily downregulated in aged mice compared to juvenile ones.

It has also been reported that re-uptake of NA by NAT in LC neurons may decrease with age (Sanders et al., 2005 [↗](#); Mitchell et al., 2017 [↗](#)). However, NAT expression in LC in 3-week-old juvenile mice was only slightly larger (by 4%) than in adult mice although the difference was significant (Mitchell et al., 2017 [↗](#)). Potent NAT activity stringently affecting autoinhibition was reported in adult rats (Williams et al., 1985 [↗](#)). Furthermore, partial inhibition of NAT in human subjects largely reduced the original level of a derivative of DOPEGAL in CSF as described above. Thus, based on the present study, it is likely that increased re-uptake of NA by NAT following excessive NA-autocrine in the absence of $\alpha 2A$ -AR, induced in response to chronic RS, resulted in the increase in active-AEP and subsequent production of tau N368. However, a further experiment to directly assess NAT activity would be necessary to firmly justify our hypothesis.

To further validate the mechanisms which we have proposed in the present study (Figure 6 [↗](#)), direct quantification of the relationship between $\alpha 2A$ -AR internalization and increased cytosolic NA levels may be desirable. However, at present, the quantification of such connection may not be possible because there appears to be no fluorescence probe to label cytosolic NA. None of NA fluorescence probes, such as NS521, BPS3 or GRAB-NE, is applicable to detect cytosolic free NA because NS521 and BPS3 anchor on plasma membrane and GRAB-NE is dependent on GPCR activation (Hettie and Glass, 2014 [↗](#); Feng et al., 2019 [↗](#); Mao et al., 2023 [↗](#)).

Conclusion

Our study unveiled the cellular mechanism for the degeneration of LC neurons under chronic RS as follows: Chronic RS impaired the autoinhibition in LC neurons by inducing Ca^{2+} - and agonist-dependent internalization of soma/dendritic $\alpha 2A$ -AR coupled with GIRK channels. This subsequently caused overexcitation/ $[Ca^{2+}]_i$ increases and increased Ca^{2+} -dependent MAO-A activity and active AEP activity, leading to a production of tau N368 fragment together with an impairment of spatial memory. These results clearly indicate that chronic RS increases cytosolic free NA to be metabolized by MAO-A, presumably by facilitating the re-uptake of autocrine-released NA due to the absence/internalization of $\alpha 2A$ -AR. Thus, in addition to NAT inhibitor, the suppression of $\alpha 2A$ -AR internalization may have a strong translational potential for developing drugs that prevent the degeneration of LC. However, the present mechanism must be proven to be valid in adult or old mice, to validate its involvement in the pathogenesis of AD.

Methods

Restraint stress

Three-weeks-old male C57BL/6 mice (JAPAN HAMAMATSU SLC, Hamamatsu, Japan) were used and housed at $24 \pm 2^\circ C$. Food and water were available ad libitum. The mice in the stressed groups were restrained in a modified 50 ml clear polystyrene conical centrifuge with multiple air holes for ventilation. The mice were subjected to 12-h restraint stress (RS; 20:00 p.m. to 8:00 a.m. of the next day) for 1, 2, 3 and 5 days. In the previous study, the duration of RS widely ranged from 0.5-h (Gray et al., 2010 [↗](#)) to 24-h (Chu et al., 2016 [↗](#)) while the RS with the duration less than 12-h has been applied daily for a period (6-h for 2 weeks (Sakaguchi and Nakamura, 1990 [↗](#)); 8-h for up to 16 weeks (Sugama et al., 2016 [↗](#)); 12-h for 2 days (Zhang et al., 2008 [↗](#))). As the prolonged stress is known to be a potential risk factor for AD (Bisht et al., 2018 [↗](#)), the protocol of daily 12-h RS is relevant for the study of functional and cytological disorders of LC neurons. Once the restraint ended, the mice were returned to their home cages with access to food and water ad libitum. The control group remained in their home cages without the restraint procedure.

Slice preparation and electrophysiological recordings

Slices were prepared from 3-weeks-old male C57BL/6 mice to investigate the effects of RS for varying periods on GIRK-I, and from C57BL/6 mice of either sex at 16–23 postnatal days (PD) to investigate the effects of $[Ca^{2+}]_i$ increases on GIRK-I, similar to our previous study (Toyoda et al., 2022). Using an Axopatch 200B (MDS Analytical Technologies, Sunnyvale, CA), whole-cell recordings were made from LC neurons that were viewed under Nomarski optics (BX51WI; Olympus, Tokyo, Japan). LC neurons were identified immunohistochemically and electrophysiologically as we previously reported (Toyoda et al., 2022). The internal solution of the patch pipettes had the following ionic composition (in mM): 123 K-gluconate, 18 KCl, 10 NaCl, 2 $MgCl_2$, 2 ATP- Na_2 , 0.3 GTP- Na_3 , 10 HEPES, and 0.2 EGTA; pH 7.3. The membrane potential values given in the text were corrected for the junction potential (10 mV) between the internal solution for the whole-cell recording (negative) and the standard extracellular solution. The pipette resistances were 4–6 M Ω . The series resistance was < 10 M Ω . All recordings were made at room temperature (RT). Records of currents and voltages were low-pass filtered at 5 kHz (3-pole Bessel filter), digitized at a sampling rate of 10 kHz (Digidata 1322A, MDS Analytical Technologies) and stored on a computer hard disk. Spike-frequency adaptation was quantitatively estimated by fitting with a saturation curve defined by Monod equation ($y = (a * x) / (b + x)$) with a saturation level (**a**) and a half saturation constant (**b**). The amplitudes of inward and outward components of NA-induced GIRK-I were measured at –130 mV and –60 mV.

Procedure for immunohistochemical examination

C57BL/6J male mice (4–5 weeks) were perfused transcardially with 5 ml of PBS, followed by 10 ml of a freshly prepared 4% paraformaldehyde in PBS. The whole brain including LC (–9 to –10 mm from bregma) was dissected out and immersed in 4% paraformaldehyde solution for 4 h at 4°C followed by cryoprotection with 30% (w/w) sucrose in PBS.

Brainstem including LC regions was cut into 8–20 μ m-thickness coronal sections on a freezing microtome (Leica Microsystems, Wetzlar, Germany). To examine the distribution of α 2A-AR in LC neurons, mouse anti-TH (1:500; sc-25269, Santa Cruz Biotechnology, Santa Cruz, CA, USA), rabbit anti-sodium potassium ATPase (1:1000; ab76020, Abcam, Cambridge, MA, USA) and goat anti- α 2A-AR (1:200; ab45871, Abcam) or rabbit anti- α 2A-AR (1:500; RA14110, Neuromics, Northfield, MN, USA) were used as primary antibodies, while cy5 donkey anti-mouse IgG (1:500; cy-2500, Vector Laboratories, Burlingame, CA, USA), FITC donkey anti-rabbit IgG (1:500; 711-095-152, Jackson ImmunoResearch, West Grove, PA, USA) and Cy3 donkey anti-goat IgG (1:500; 706-165-147, Jackson ImmunoResearch) were used as secondary antibodies. We used the affinity purified IgG antibody for α 2A-AR immunohistochemistry, in which antibody specificity was ensured as examined previously (Stone et al., 1998; Tan et al., 2009). To examine the expression in LC neurons, mouse anti-TH (1:500) and rabbit anti-MAO-A (1:2000; ab126751, Abcam) were used as primary antibodies, while FITC donkey anti-mouse IgG (1:500; 715-545-150; Jackson ImmunoResearch) for TH and Cy3 donkey anti-rabbit IgG (1:2000; 711-165-152; Jackson ImmunoResearch) were used as secondary antibodies. To examine the distribution of legumain in LC neurons, mouse anti-legumain (1:200; sc-133234, Santa Cruz) and rabbit anti-DBH (1:500; 22806, ImmunoStar, Hudson, WI, USA) were used as primary antibodies, while FITC donkey anti-mouse IgG (1:500; 715-545-150, Jackson ImmunoResearch) and Cy3 donkey anti-rabbit IgG (1:500; 711-165-152, Jackson ImmunoResearch) were used as secondary antibodies. The slices were coverslipped with mounting media (H1000, Vector Laboratories), and sections were visualized with a confocal microscope (LSM980, Carl Zeiss, Oberkochen, Germany) at 10x and 63x magnification.

Western blot study

Brain slices from the control and 5-day RS-treated mice were sectioned with vibratome (VT1000P, Leica Microsystems) to obtain 500 μm -thickness coronal sections containing LC region (bregma – 5.34 mm to –5.80 mm). We performed western blot study only in 5-day RS mice without examining in 3-day RS mice, under the assumption that for the degeneration-related biochemical changes to reach sufficiently detectable levels, a high $[\text{Ca}^{2+}]_i$ condition due to the rundown of GIRK-I have to be sustained for some period of time after the rundown of GIRK-I reached a plateau level in 3-day RS mice (see **Figure 4F**, G). LC dissection was done via micro punch using 1.5 mm biopsy punch (BP-15F, Kai medical, Gifu, Japan) bilaterally under light microscope. Dissected LC were quickly frozen in liquid nitrogen and stored at -80°C until processed. Because LC region was too small to collect sufficient tissue from one mouse for the analysis, the punched LC tissues were collected from two to three mice to obtain a protein lysate sample for each western blotting analysis. LC tissues were homogenized by bead homogenizer in T-PER™ Tissue Protein Extraction Reagent (78510, Thermo Fisher) containing protease and phosphatase inhibitor cocktails (MilliporeSigma, St. Louis, MO, USA). The membrane and cytosol proteins from LC lysate were isolated using Mem-PER Plus Kit (78840, Thermo Fisher) in accordance with the manufacture's protocol. The protein concentration of the lysate was determined using a Lowry protein assay (Bio-Rad, Hercules, CA, USA). Protein samples (20 μg) were heated to 95°C for 5 min, separated by 12% sodium dodecyl sulfate-polyacrylamide gel electrophoresis, and transferred to a polyvinylidene fluoride membrane using a Transblot SD apparatus (Bio-Rad). After the blots had been washed with tris-buffered saline containing Tween-20 (TBST, 10 mM Tris-HCl, 150 mM NaCl, 0.05% Tween-20), the membranes were blocked with 2% BSA and 4% skim milk/TBST at RT for 1 h and incubated at 4°C overnight with primary antibodies, followed by an anti-rabbit IgG horseradish peroxidase (HRP)-conjugated secondary antibody (1:5,000, 7074; Cell Signaling Technology, Danvers, MA, USA) at RT for 1 h. The western blot membrane was stripped the primary and secondary antibodies and reprobed with the anti- β -actin antibody followed by the anti-rabbit IgG HRP-conjugated secondary antibody. The following primary antibodies were used for the blots: rabbit anti- $\alpha 2\text{A-AR}$ (1:500; AAR020, alomone), rabbit anti-GAPDH (1:1000; LF-PA0018, Abfrontier, Seoul, Korea), mouse anti-beta actin (1:2000, A5441, MilliporeSigma) and rabbit anti-sodium potassium ATPase (1:1000; ab76020, Abchem) (**Figure 3C, F and G**); rabbit anti-TH (1:1600; GTX113016, GeneTex, Irvine, CA, USA), rabbit anti-MAO-A (1:6400; ab126751, Abcam), mouse anti-legumain (AEP) (1:400; sc-133234, Santa Cruz), mouse anti-TAU-5 (1:1500, Santa Cruz Biotechnology, SC-58860) and mouse anti-beta actin (1:12800, 66009-1-Ig, Proteintec, Rosemont, IL, USA) (**Figure 5A, D and G**); rabbit anti-tau AEP-cleaved (N368) (1:5000, ABN1703, MilliporeSigma) and mouse anti-beta actin (1:1000, A5441, MilliporeSigma) (**Figure 5G, I**). Western blot analysis was performed by measuring band intensity with ImageJ 1.50i software (National Institutes of Health, Bethesda, MD, USA).

Quantitative real-time PCR

cDNA was synthesized with Super Script II Reverse Transcriptase (18064, Invitrogen, Carlsbad, CA, USA) with 1 mg of LC neuron RNA. Using a 7300 Gene Amp PCR system (Applied Biosystems, Carlsbad), quantitative real-time PCR was run in Power SYBR Green PCR master mix (4367656; Life Technologies, Carlsbad) under the following thermal cycling conditions; an initial step of $94^\circ\text{C}/5$ min, followed by 40 cycles of a series of steps of $94^\circ\text{C}/45$ s, $61^\circ\text{C}/45$ s, $72^\circ\text{C}/45$ s and ended with a series of steps of $95^\circ\text{C}/60$ s, $60^\circ\text{C}/1$ min, $95^\circ\text{C}/15$ s, $60^\circ\text{C}/15$ s. The following primers were used: $\alpha 2\text{A}$ (forward GGAATCATGGCTGTGGAGAT, reverse CAGAAATCCCTTCCCTGTCA); $\alpha 2\text{C}$ (forward TCATCGTTTTCACCGTGGTA, reverse GCTCATTGGCCAGAGAAAAG); GIRK1 (forward ACCCTGGTGGATCTCAAGTG, reverse GGCCACACAGGGAGTGTAGT); GRIK2 (forward CAACCTCAACGGGTTTGTCT, reverse ATCCTACCATGAAGGCGTTG); GIRK3 (forward CGTCTCACCTCTCGTCATCA, reverse CATCCACCAGGTACGAGCTT); GAPDH (forward CCAGAACATCATCCCTGCAT, reverse GCATCGAAGGTGGAAGAGTG).

Behavioral test

The control mice as non-stress group and mice exposed to RS for 1 day, 3 days and 5 days as 1-day, 3-day and 5-day RS groups were tested on the Y maze apparatus, consisted of three identical arms in a “Y” shape made of white acrylic (35 cm in length, 5 cm in width, and 20 cm in height) to assess working spatial memory. All tests were performed between 9 AM and noon. Investigators were blinded to the group when performing experiments. Each mouse was placed at the extremity of the “start arm” and left for a training session of 15 minutes, with one of the 2 other arms closed with a white acrylic board (the closed arm was randomly chosen for each mouse). Then, the mouse was brought back to its cage for 5 minutes, the maze was cleaned, the closed arm was opened, and the mouse placed at the extremity of the start arm for a 5-minute test session, under video tracking. The time spent and distance traveled in each arm (start, familiar, and novel arms) were measured and analyzed with Smart 3.0 video tracking software (Panlab, Harvard Apparatus, Holliston, MA, USA). Given their natural curiosity, mice are expected to spend more time exploring the new arm.

Drug application

Atipamezole as a blocker of $\alpha 2$ -AR and tertiapin-Q as a GIRK blocker were bath-applied at 5-10 μ M and 200 nM, respectively. Barbadin as an arrestin/AP2 blocker was added to the internal solution at a final concentration of 100 μ M (Beautrais et al., 2017 [DOI](#)). All chemicals except tertiapin-Q were purchased from MilliporeSigma unless specified otherwise. Tertiapin-Q was purchased from Tocris Bioscience (Bristol, UK).

Statistical Analysis

Statistical analysis was performed using STATISTICA10J (StatSoft Japan, Tokyo, Japan). Numerical data were expressed as the mean $\pm$ SD. The statistical significance was assessed using Student's *t*-test, one-way ANOVA followed by post hoc Fisher's PLSD and Wilk's lambda. The significance level was set at 5% or less ($p < 0.05$). Statistical results are given as a *p* value, unless $p < 0.001$. All the statistical data contained in this study are presented in Table S1.

Data Availability

All data and/or analyses generated during the current study are available from the corresponding author upon request.

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Additional information

Author Contributions

YK, HT, and SBO conceived and designed the study. HT, DK, BGK, and TS performed experiments. HT, DK, BGK, TS, TK, and YK analyzed data. YK, HT and SBO wrote the manuscript.

Ethics

This study was conducted following the Helsinki Declaration, and carried out according to the ARRIVE guidelines. All experimental procedures were approved and performed in accordance with the relevant guidelines of the Ethical Guidelines Committee at Osaka University, Seoul National University and Kyushu University.

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Additional files

Supplemental Figures 

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Reviewer #1 (Public review):

This study aims to elucidate the mechanisms by which stress-induced $\alpha 2A$ -adrenergic receptor ($\alpha 2A$ -AR) internalization leads to cytosolic noradrenaline (NA) accumulation and subsequent neuronal dysfunction in the locus coeruleus (LC). While the manuscript presents an interesting but ambitious model involving calcium dynamics, GIRK channel rundown, and autocrine NA signaling, several key limitations undermine the strength of the conclusions.

First, the revision does not include new experiments requested by reviewers to validate core aspects of the mechanism. Specifically, there is no direct measurement of cytosolic NA levels or MAO-A enzymatic activity to support the link between receptor internalization and neurochemical changes. The authors argue that such measurements are either not feasible or beyond the scope of the study, leaving a significant gap in the mechanistic chain of evidence.

Second, the behavioral analysis remains insufficient to support claims of cognitive impairment. The use of a single working memory test following an anxiety test is inadequate to verify memory dysfunction behaviors. Additional cognitive assays, such as the Morris Water Maze or Novel Object Recognition, are recommended but not performed.

Third, concerns regarding the lack of rigor in differential MAO-A expression in fluorescence imaging were not addressed experimentally. Instead of clarifying the issue, the authors moved the figure to supplementary data without providing further evidence (e.g., an

enzymatic assay or quantitative reanalysis of Western blot, or re-staining of IF for MAO-A) to support their interpretation.

Fourth, concerns regarding TH staining remain unresolved. In Figure S7, the α 2A-AR signal appears to resemble TH staining, and vice versa, raising the possibility of labeling errors. It is recommended that the authors re-examine this issue by either double-checking the raw data or repeating the immunostaining to validate the staining.

Overall, the manuscript offers a potentially interesting framework but falls short in providing the experimental rigor necessary to establish causality. The reliance on indirect reasoning and reorganizing of existing data, rather than generating new evidence, limits the overall impact and interpretability of the study.

<https://doi.org/10.7554/eLife.106362.2.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript investigates the mechanism by which chronic stress induces degeneration of locus coeruleus (LC) neurons. The authors demonstrate that chronic stress leads to the internalization of α 2A-adrenergic receptors (α 2A-ARs) on LC neurons, causing increased cytosolic noradrenaline (NA) accumulation and subsequent production of the neurotoxic metabolite DOPEGAL via monoamine oxidase A (MAO-A). The study suggests a mechanistic link between stress-induced α 2A-AR internalization, disrupted autoinhibition, elevated NA metabolism, activation of asparagine endopeptidase (AEP), and Tau pathology relevant to Alzheimer's disease (AD). The conclusions of this paper are largely well-supported by the data, but some aspects of image acquisition require further examination.

Strengths:

This study clearly demonstrates the effects of chronic stimulation on the excitability of LC neurons using electrophysiological techniques. It also elucidates the role of α 2-adrenergic receptor (α 2-AR) internalization and the associated upstream and downstream signaling pathways of GIRK-1, using a range of pharmacological agents, highlighting the innovative nature of the work. Additionally, the study identifies the involvement of the MAO-A-DOPEGAL-AEP pathway in this process. The topic is timely, the proposed mechanistic pathway is compelling, and the findings have translational relevance, particularly about therapeutic strategies targeting α 2A-AR internalization in neurodegenerative diseases.

Weaknesses:

- (1) The manuscript reports that chronic stress for 5 days increases MAO-A levels in LC neurons, leading to the production of DOPEGAL, activation of AEP, and subsequent tau cleavage into the tau N368 fragment, ultimately contributing to neuronal damage. However, the authors used wild-type C57BL/6 mice, and previous literature has indicated that AEP-mediated tau cleavage in wild-type mice is minimal and generally insufficient to cause significant behavioral alterations. Please clarify and discuss this apparent discrepancy.
- (2) It is recommended that the authors include additional experiments to examine the effects of different durations and intensities of stress on MAO-A expression and AEP activity. This would strengthen the understanding of stress-induced biochemical changes and their thresholds.
- (3) Please clarify the rationale for the inconsistent stress durations used across Figures 3, 4, and 5. In some cases, a 3-day stress protocol is used, while in others, a 5-day protocol is

applied. This discrepancy should be addressed to ensure clarity and experimental consistency.

(4) The abbreviation "vMAT2" is incorrectly formatted. It should be "VMAT2," and the full name (vesicular monoamine transporter 2) should be provided at first mention.

Comments on revisions:

The authors have addressed all of the reviewers' comments.

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Reviewer #3 (Public review):

Summary:

The authors present a technically impressive data set showing that repeated excitation or restraint stress internalises somatodendritic $\alpha 2A$ adrenergic autoreceptors ($\alpha 2A$ ARs) in locus coeruleus (LC) neurons. Loss of these receptors weakens GIRK-dependent autoinhibition, raises neuronal excitability, and is accompanied by higher MAO A, DOPEGAL, AEP, and tau N368 levels. The work combines rigorous whole-cell electrophysiology with barbadin-based trafficking assays, qPCR, Western blotting and immunohistochemistry. The final schematic is appealing and in principle, could explain early LC hyperactivity followed by degeneration in ageing and Alzheimer's disease.

Strengths:

Multi-level approach - The study integrates electrophysiology, pharmacology, mRNA quantification, and protein-level analysis.

Use of barbadin to block β -arrestin/AP-2-dependent internalisation is both technically precise and mechanistically informative

Well-executed electrophysiology

translation relevance

converges to a model that peers discussed (scientists can only discuss models - not data!)

Weaknesses:

Nevertheless, the manuscript currently reads as a sequence of discrete experiments rather than a single causal chain

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Weaknesses:

(1) *The manuscript's logical flow is challenging and hard to follow, and key arguments could be more clearly structured, particularly in transitions between mechanistic components.*

We have revised our manuscript so as to make it easy for readers to follow the logical flow in transitions between mechanistic components by adding the descriptions of Figure S1E-J, Figure S2F-K, Figure S3A-H, Figure S4A-F, Figure S5, and Figure S6 in the revised manuscript.

(2) The causality between stress-induced $\alpha 2A$ -AR internalization and the enhanced MAO-A remains unclear. Direct experimental evidence is needed to determine whether $\alpha 2A$ -AR internalization itself or Ca^{2+} drives MAO-A activation, and how they activate MAO-A should be considered.

We believe that the causality between stress-induced $\alpha 2A$ -AR internalization and the enhancement of MAO-A is clearly demonstrated by our current experiments, while our explanations may be improved by making them easier to understand especially for those who are not expert on electrophysiology.

Firstly, it is well established that autoinhibition in LC neurons is mediated by $\alpha 2A$ -AR coupled-GIRK (Arima et al., 1998, J Physiol; Williams et al., 1985, Neuroscience). We found that spike frequency adaptation in LC neurons was also mediated by $\alpha 2A$ -AR coupled GIRK-I (Figure 1A-I), and that $\alpha 2A$ -AR coupled GIRK-I underwent $[Ca^{2+}]_i$ dependent rundown (Figures 2, S1, S2), leading to an abolishment of spike-frequency adaptation (Figures S4). $[Ca^{2+}]_i$ dependent rundown of $\alpha 2A$ -AR coupled GIRK-I was prevented by barbadin (Figure 2G-J), which prevents the internalization of G-protein coupled receptor (GPCR) channels.

Abolishment of spike frequency adaptation itself, i.e., “increased spike activity” can increase $[Ca^{2+}]_i$ because $[Ca^{2+}]_i$ is entirely dependent on the spike activity as shown by $[Ca^{2+}]_i$ imaging method in Figure S3.

Thus, $\alpha 2A$ -AR internalization can increase $[Ca^{2+}]_i$ through the abolishment of autoinhibition or spike frequency adaptation, and a $[Ca^{2+}]_i$ increase drives MAO-A activation as reported previously (Cao et al., 2007, BMC Neurosci). The mechanism how Ca^{2+} activates MAO-A is beyond the scope of the current study.

Our study just focused on the mechanism how chronic or sever stress can cause persistent overexcitation and how it results in LC degeneration.

(3) The connection between $\alpha 2A$ -AR internalization and increased cytosolic NA levels lacks direct quantification, which is necessary to validate the proposed mechanism.

Direct quantification of the relationship between $\alpha 2A$ -AR internalization and increased cytosolic NA levels may not be possible, and may not be necessarily needed to be demonstrated as explained below.

The internalization of $\alpha 2A$ -AR can increase $[Ca^{2+}]_i$ through the abolishment of autoinhibition or spike frequency adaptation, and $[Ca^{2+}]_i$ increases can facilitate NA autocrine (Huang et al., 2007), similar to the transmitter release from nerve terminals (Kaesler & Regehr, 2014, Annu Rev Physiol).

Autocrine released NA must be re-uptaken by NAT (NA transporter), which is firmly established (Torres et al., 2003, Nat Rev Neurosci). Re-uptake of NA by NAT is the only source of intracellular NA, and NA re-uptake by NAT should be increased as the internalization of NA binding site ($\alpha 2A$ -AR) progresses in association with $[Ca^{2+}]_i$ increases (see page 11, lines 334-336).

Thus, the connection between $\alpha 2A$ -AR internalization and increased cytosolic NA levels is logically compelling, and the quantification of such connection may not be possible at present (see the response to the comment made by the Reviewer #1 as Recommendations for the authors (2) and beyond the scope of our current study.

(4) The chronic stress model needs further validation, including measurements of stress-induced physiological changes (e.g., corticosterone levels) to rule out systemic effects that may influence LC activity. Additional behavioral assays for spatial memory impairment should also be included, as a single behavioral test is insufficient to confirm memory dysfunction.

It is well established that restraint stress (RS) increases corticosterone levels depending on the period of RS (García-Iglesias et al., 2014, Neuropharmacology), although we are not reluctant to measure the corticosterone levels. In addition, there are numerous reports that showed the increased activity of LC neurons in response to various stresses (Valentino et al., 1983; Valentino and Foote, 1988; Valentino et al., 2001; McCall et al., 2015), as described in the text (page 4, lines 96-98). Measurement of cortisol levels may not be able to rule out systemic effects of CRS on the whole brain.

We had already done another behavioral test using elevated plus maze (EPM) test. By combining the two tests, it may be possible to more accurately evaluate the results of Y-maze test by differentiating the memory impairment from anxiety. However, the results obtained by these behavioral tests are just supplementary to our current aim to elucidate the cellular mechanisms for the accumulation of cytosolic free NA. Therefore, we have softened the implication of anxiety and memory impairment (page 13, lines 397-400 in the revised manuscript).

(5) Beyond b-arrestin binding, the role of alternative internalization pathways (e.g., phosphorylation, ubiquitination) in α2A-AR desensitization should be considered, as current evidence is insufficient to establish a purely Ca²⁺-dependent mechanism.

We can hardly agree with this comment.

It was clearly demonstrated that repeated application of NA itself did not cause desensitization of α2A-AR (Figure S1A-D), and that the blockade of b-arrestin binding by barbadin completely suppressed the Ca²⁺-dependent downregulation of GIRK (Figure 2G-K). These observations can clearly rule out the possible involvement of phosphorylation or ubiquitination for the desensitization.

Not only the barbadin experiment, but also the immunohistochemistry and western blot method clearly demonstrated the decrease of α2A-AR expression on the cell membrane (Figure 3).

Ca²⁺-dependent mechanism of the rundown of GIRK was convincingly demonstrated by a set of different protocols of voltage-clamp study, in which Ca²⁺ influx was differentially increased. The rundown of GIRK-I was orderly potentiated or accelerated by increasing the number of positive command pulses each of which induces Ca²⁺ influx (compare Figure S1E-J, Figure S2A-E and Figure S2F-K along with Figure 2A-F). The presence or absence of Ca²⁺ currents and the amount of Ca²⁺ currents determined the trend of the rundown of GIRK-I (Figures 2, S1 and S2). Because the same voltage protocol hardly caused the rundown when it did not induce Ca²⁺ currents in the absence of TEA (Figure S1F; compare with Figure 2B), blockade of Ca²⁺ currents by nifedipine would not be so beneficial.

We believe the series of voltage-clamp protocols convincingly demonstrated the orderly involvement of [Ca²⁺]_i in accelerating the rundown of GIRK-I.

(6) NA leakage for free NA accumulation is also influenced by NAT or VMAT2. Please discuss the potential role of VMAT2 in NA accumulation within the LC in AD.

It has been demonstrated that reduced VMAT2 levels increased susceptibility to neuronal damage: VMAT2 heterozygote mice displayed increased vulnerability to MPTP as evidenced by reductions in nigral dopamine cell counts (Takahashi et al, 1997, PNAS). Thus, when the activity of VMAT2 in LC neurons were impaired by chronic restraint stress, cytosolic NA levels in LC neurons would increase. We have added such discussion in the revised manuscript (page 12, lines 381-384).

(7) Since the LC is a small brain region, proper staining is required to differentiate it from surrounding areas. Please provide a detailed explanation of the methodology used to define LC regions and how LC neurons were selected among different cell types in brain slices for whole-cell recordings.

LC neurons were identified immunohistochemically and electrophysiologically as we previously reported (see Fig. 2 in Front. Cell. Neurosci. 16:841239. doi: 10.3389/fncel.2022.841239). We have added this explanation in the method section of the revised manuscript (page 15, lines 474-475). A delayed spiking pattern in response to depolarizing pulses (Figure S10 in the revised manuscript) applied at a hyperpolarized membrane potential was commonly observed in LC neurons in many studies (Masuko et al., 1986; van den Pol et al., 2002; Wagner-Altendorf et al., 2019).

Reviewer #2 (Public review):

Weaknesses:

(1) The manuscript reports that chronic stress for 5 days increases MAO-A levels in LC neurons, leading to the production of DOPEGAL, activation of AEP, and subsequent tau cleavage into the tau N368 fragment, ultimately contributing to neuronal damage. However, the authors used wild-type C57BL/6 mice, and previous literature has indicated that AEP-mediated tau cleavage in wild-type mice is minimal and generally insufficient to cause significant behavioral alterations. Please clarify and discuss this apparent discrepancy.

In our study, normalized relative value of AEP-mediated tau cleavage (Tau N368) was much higher in CRS mice than non-stress wild-type mice. It is not possible to compare AEP-mediated tau cleavage between our non-stress wild type mice and those observed in previous study (Zhang et al., 2014, Nat Med), because band intensity is largely dependent on the exposure time and its numerical value is the normalized relative value. In view of such differences, our apparent band expression might have been intensified to detect small changes.

(2) It is recommended that the authors include additional experiments to examine the effects of different durations and intensities of stress on MAO-A expression and AEP activity. This would strengthen the understanding of stress-induced biochemical changes and their thresholds.

GIRK rundown was almost saturated after 3-day RS and remained the same in 5-day RS mice (Fig. 4A-G), which is consistent with the downregulation of $\alpha 2A$ -AR and GIRK1 expression by 3-day RS (Fig. 3C, F and G; Fig. 4J and K). However, we examined the protein levels of MAO-A, pro/active-AEP and Tau N368 only in 5-day RS mice without examining in 3-day RS mice. This is because we considered the possibility that a high $[Ca^{2+}]_i$ condition may have to be sustained for some period of time to induce changes in MAO-A, AEP and Tau N368, and therefore 3-day RS may be insufficient to induce such changes. We have added this in the revised manuscript (page 17, lines 521-525).

(3) Please clarify the rationale for the inconsistent stress durations used across Figures 3, 4, and 5. In some cases, a 3-day stress protocol is used, while in others, a 5-day protocol is applied. This discrepancy should be addressed to ensure clarity and experimental consistency.

Please see our response to the comment (2).

(4) The abbreviation "vMAT2" is incorrectly formatted. It should be "VMAT2," and the full name (vesicular monoamine transporter 2) should be provided at first mention.

Thank you for your suggestion. We have revised accordingly.

Reviewer #3 (Public review):

Weaknesses:

Nevertheless, the manuscript currently reads as a sequence of discrete experiments rather than a single causal chain. Below, I outline the key points that should be addressed to make the model convincing.

Please see the responses to the recommendation for the authors made by reviewer #3.

Reviewer #1 (Recommendations for the authors):

(1) Improve the clarity and organization of the manuscript, ensuring smoother transitions between concepts and mechanisms.

Please see the response to the comment raised by Reviewer #1 as Weakness

(2) Adjust any quantifying method for cytosolic NA levels under different conditions to support the link between receptor internalization and NA accumulation.

If fluorescent indicator of cytosolic free NA is available, it would be possible to measure changes in cytosolic NA levels. However, at present, there appeared to be no fluorescence probe to label cytosolic NA. For example, NS521 labels both dopamine and norepinephrine inside neurosecretory vesicles (Hettie & Glass et al., 2014, Chemistry), and BPS3 fluorescence sensor labels NA around cell membrane by anchoring on the cell membrane (Mao et al., 2023, Nat Comm). Furthermore, the method reported in “A Genetically Encoded Fluorescent Sensor for Rapid and Specific In Vivo Detection of Norepinephrine” is limited to detect NA only when $\alpha 2AR$ is expressed. In the present study, increases in cytosolic NA levels are caused by internalization of $\alpha 2AR$. Cytosolic NA measurements with GRAB NE photometry may not be applicable in the present study. However, we have discussed the availability of such fluorescent methods to directly prove the increase in cytosolic NA as a limitation of our study (page 14, lines 429-436 in the revised manuscript).

(3) Include validation of the chronic stress model with physiological and behavioral measures (e.g., corticosterone levels and another behavioral test).

Please see the response to the comment raised by Reviewer #1 as Weakness (4).

(4) All supplemental figures should be explicitly explained in the Results section. Specifically, clarify and describe the details of Figure S1G-K, Figure S2F-K, Figure S3A-H, Figure S4A-F, Figure S5, and Figure S6 to ensure all supplementary data are fully integrated into the main text.

We have more explicitly and clearly described the details of Figure S1E-J, Figure S2F-K, Figure S3A-H, Figure S4A-F, Figure S5, and Figure S6 and fully integrated those explanations into the main text in the revised manuscript.

(5) In Figure 3, the morphology of TH-positive cells differs between panels D and E. Additionally, TH is typically expressed in the cytosol, but in the provided images, it appears to be localized only to the membrane. Please clarify this discrepancy and provide a lower-magnification image to display a larger area, not one cell.

In a confocal image, TH is not necessarily expressed homogeneously in the cytosol, but is expressed in a ring-shaped pattern inside the plasma membrane, avoiding the cell nucleus and its surrounding Golgi apparatus and endoplasmic reticulum (ER) (Henrich et al., 2018, *Acta Neuropathol Commun*; see Fig. 4a and 6e), especially when the number of z-stack of confocal images is small. This is presumably because LC neurons are especially enriched with numerous Golgi apparatus and ER (Groves & Wilson, 1980, *J Comp Neurol*).

In Figure S7, we showed a lower-magnification image of LC and its adjacent area (mesencephalic trigeminal nucleus). In the LC area, there are a variety of LC neurons, which include oval shaped neurons (open arrowhead; similar to Figure 3D) and also rhombus-like shaped neurons (open double arrowheads, similar to Figure 3E). A much lower-magnification image of LC neurons constituting LC nucleus was shown in Figure 5A.

(6) In Figure 5, the difference in MAO-A expression is not clearly visible in the fluorescence images. Enzymatic assays for AEP and MAO-A should be included to demonstrate the increased activity better.

In the current study, we did not elaborate to detect the changes in TH, MAO-A and AEP in terms of immunohistochemical method. Instead, we elaborated to detect such changes in terms of western blot method. The main conclusions in the current study were drawn primarily by electrophysiological techniques as we have expended much effort on electrophysiological experiments. Because the relative quantification of active AEP and Tau N368 proteins by western blotting analysis may accurately reflect changes in those enzyme activities, enzymatic assay may not be necessarily required but is helpful to better demonstrate AEP and MAO-A activity. We have described the necessity of enzymatic assay to better demonstrate the AEP and MAO-A activities (page 10, lines 314-315).

Reviewer #3 (Recommendations for the authors):

(1) Causality across the pathway

Each step ($\alpha 2A$ internalisation, GIRK rundown, Ca^{2+} rise, MAO-A/AEP upregulation) is demonstrated separately, but no experiment links them in a single preparation. Consider in vivo Ca^{2+} or GRAB NE photometry during restraint stress while probing $\alpha 2A$ levels with i.p. clonidine injection or optogenetic over excitation coupled to biochemical readouts. Such integrated evidence would help to overcome the correlational nature of the manuscript to a more mechanistic study.

It is not possible to measure free cytosolic NA levels with GRAB NE photometry when $\alpha 2A$ AR is internalized as described above (see the response to the comment made by reviewer #1 as the recommendation for the authors).

(2) Pharmacology and NE concentration

The use of 100 μM noradrenaline saturates α and β adrenergic receptors alike. Please provide ramp measurements of GIRK current in dose-response at 1-10 μM NE (blocked by

atipamezole) to confirm that the rundown really reflects $\alpha 2A$ activity rather than mixed receptor effects.

It is true that 100 μM noradrenaline activates both α and β adrenergic receptors alike. However, it was clearly showed that enhancement of GIRK-I by 100 μM noradrenaline was completely antagonized by 10 μM atipamezole and the Ca^{2+} dependent rundown of NA-induced GIRK-I was prevented by 10 μM atipamezole. Considering the K_i values of atipamezole for $\alpha 2A$ AR ($\approx 1\sim 3$ nM) (Vacher et al., 2010, J Med Chem) and β AR (> 10 μM) (Virtanen et al., 1989, Arch Int Pharmacodyn Ther), these results really reflect $\alpha 2A$ AR activity but not β AR activity (Figure S5). Furthermore, because it is already well established that NA-induced GIRK-I was mediated by $\alpha 2A$ AR activity in LC neurons (Arima et al., 1998, J Physiol; Williams et al., 1985, Neuroscience), it is not necessarily need to re-examine 1-10 μM NA on GIRK-I.

(3) Calcium dependence is not yet definitive

The rundown is induced with a TEA-enhanced pulse protocol. Blocking L-type channels with nifedipine (or using Cd^{2+}) during this protocol should show whether Ca^{2+} entry is necessary. Without such a control, the Ca^{2+} link remains inferential.

The Ca^{2+} link was precisely demonstrated by a series of voltage clamp experiment, in which Ca^{2+} influx was orderly potentiated by increasing the number of positive voltage pulses (Figures S1 and S2). As the number of positive voltage pulses was increased, the rundown of GIRK-I was accelerated or enhanced more. The relationship between the number of spikes and the Ca^{2+} influx detected as Ca^{2+} transients was well documented in Ca^{2+} imaging experiments using fura-2 (Figure S3).

The presence or absence of Ca^{2+} currents and the amount of Ca^{2+} currents determined the trend of the rundown of GIRK-I (Figs. 2, S1 and S2). The same voltage protocol hardly caused the rundown when it did not induce Ca^{2+} currents in the absence of TEA (Fig. S1F; compare with Fig. 2B), and the series of voltage-clamp protocols convincingly demonstrated the orderly involvement of $[\text{Ca}^{2+}]_i$ in accelerating the rundown of GIRK-I. Therefore, blockade of Ca^{2+} currents by nifedipine may not be so beneficial.

(4) Age mismatch and disease claims

All electrophysiology and biochemical data come from juvenile ($< P30$) mice, yet the conclusions stress Alzheimer-related degeneration. Key endpoints need to be replicated in adult or aged mice, or the manuscript should soften its neurodegenerative scope.

As described in the section of Conclusion, we never stress Alzheimer-related degeneration, but might give such an impression. To avoid such a misunderstanding, we have added a description “However, the present mechanism must be proven to be valid in adult or old mice, to validate its involvement in the pathogenesis of AD.” (page 14, lines 448-450).

(5) Direct evidence for extracellular/cytosolic NE

The proposed rise in reuptake NA is inferred from electrophysiology. Modern fluorescent sensors (GRAB NE, nLight) or fast scan voltammetry could quantify NE overflow and clearance during stress, directly testing the model.

Please see the response to the comment made by Reviewer #1 as the Recommendations for the authors (2) as described above.

(6) Quantitative histology

Figure 5 presents attractive images but no numerical analysis. Please provide ROI-based fluorescence quantification (with n values) or move the images to the supplement and rely on the Western blots.

We have moved the immunohistochemical results in Fig. 5 to the supplement as we believe the quantification of immunohistochemical staining is not necessarily correct.

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